

AI for Environmental Monitoring Project

Problem statement:

Traditional environmental monitoring methods lack the speed and resolution needed to respond to urgent environmental threats. Critical Issues such as air and water pollution, climate change, and deforestation require smarter and faster monitoring systems. This project explores how AI, specifically deep learning and computer vision, can improve the accuracy and scalability of environmental monitoring by evaluating five well known AI methods.

Methods:

- Convolutional Neural Networks (CNNs) for image classification monitoring and other tasks that involve spatial data (e.g., detecting deforestation, wildfire smoke, or algal blooms from satellite imagery).
- Recurrent Neural Networks (RNNs) and LSTMs for analyzing time series environmental sensor data (e.g., air quality, temperature, predict flood events, rainfall data, or pollutant levels).
- Support Vector Machines (SVMs) supervised learning for detecting unsafe environmental conditions. (e.g., clean or polluted water)
- Unsupervised anomaly detection for spotting unusual or rare environmental events (e.g., K-means clustering to identify chemical leaks or illegal waste dumping).
- Random Forests with the use of decision trees for classification or regression (e.g., predicting pollution levels, water quality, or soil conditions)

Experimental Setup: We will use a combination of publicly available datasets, relevant evaluation metrics, and common machine learning frameworks.

Datasets used for Evaluation:

- Kaggle Water Quality Dataset – provides a physical/chemical properties of water that determine if it is potable or not.
- Wildfire Open Datasets – Fire locations, durations and severity. (worldwide)
- NOAA Global Historical Climate Network (GHCN) – Daily weather data from stations worldwide.
- Sentinel-2 (ESA) – Satellite Imagery for visual monitoring.
- Landsat 8 – US satellite imagery for detecting water, fire, land cover.

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- Planet Disaster Dataset - satellite imagery for disasters like floods or hurricanes, etc.
- UCI Air Quality Dataset – hourly measurements of air pollutants in Italy.
- Environmental Anomaly Data - Synthetic or real world datasets with labeled anomalies (e.g., chemical leaks or illegal dumping) for unsupervised anomaly detection experiments.

Evaluation metrics will include:

- Accuracy, Precision, Recall, and F1 Score for prediction evaluation, and Intersection over Union (IoU) for image based classification/detection tasks.
- Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) and R squared (R^2) for regression or forecasting.

Tools: We plan to use Python with libraries such as TensorFlow, PyTorch for deep learning models(CNNs, ViTs, RNNs/LSTMs, Multimodal learning), scikit-learn (SVMs, Decision Trees, Random Forests, K-means), and possibly Google Earth Engine for geospatial data processing. Jupyter Notebooks or Google Collab for code documentation.